

Temperature Sensing and Max-Selection Circuit Design

Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Design Objective

The slave board monitors six cell temperatures using NTC thermistors, but the LTC6811 auxiliary input resources are limited. To preserve safety coverage, thermistors are grouped and passed through an analog selection circuit that reports the hottest cell in each group.

- Sensor type: NTCALUG91A103GLA thermistor, 10 kOhm at 25 C, +/-2% tolerance.
- Safety principle: report the worst-case cell temperature rather than averaging cells.
- Verification target: cell temperature measurement error $\leq \pm 4$ C compared with a thermocouple reference.

NTC Divider Equation

Each thermistor is used in a voltage divider. With the thermistor referenced to ground, the measured node voltage is proportional to the thermistor resistance.

$$V_{\text{node}} = V_{\text{ref}} \cdot R_{\text{NTC}} / (R_{\text{fixed}} + R_{\text{NTC}})$$

For an NTC device, resistance decreases as temperature increases. Therefore the node voltage decreases as the cell temperature increases.

$$T \text{ increases} \Rightarrow R_{\text{NTC}} \text{ decreases} \Rightarrow V_{\text{node}} \text{ decreases}$$

Thermistor Resistance Model

The thermistor resistance can be modeled with the beta equation. Temperature values are in Kelvin for this equation.

$$R_{\text{NTC}}(T) = R_0 \cdot \exp[\beta \cdot (1/T - 1/T_0)]$$

Symbol	Meaning
$R_{\text{NTC}}(T)$	Thermistor resistance at absolute temperature T .
R_0	Nominal thermistor resistance at reference temperature T_0 , typically 10 kOhm at 25 C.
β	Thermistor material constant from the component datasheet.
T, T_0	Absolute temperatures in Kelvin.

Selection Logic

Since higher temperature produces lower divider voltage, the circuit must select the minimum voltage to represent the maximum temperature. For a group with three thermistor voltages V_A , V_B , and V_C :

$$V_{\text{out}} = \min(V_A, V_B, V_C)$$

$$T_{\text{reported}} = \max(T_1, T_2, T_3)$$

This mapping is conservative. If one cell in the group becomes hot, the selected output follows that cell rather than hiding the event through averaging.

Active Minimum-Selection Circuit

The circuit uses TL974 op-amp stages with diodes in feedback. Placing the diode inside the feedback loop creates an active-rectifier behavior, which compensates for diode forward voltage and allows the selected bus voltage to track the lowest input accurately.

Stage	Function	Technical Reason
Active rectification	Op-amp and Schottky diode track the input while compensating diode drop.	Avoids the normal diode forward-voltage error in a passive diode-OR circuit.
Common pull-up bus	A 25 kOhm pull-up biases the bus high until one channel sinks it.	Lets the lowest-voltage channel dominate the output.
Diode blocking	Non-selected channels are isolated when another input clamps the bus.	Prevents higher-voltage thermistor channels from corrupting the selected minimum.

Verification Result

The sensing chain was evaluated by comparing LTC6811-reported temperature to thermocouple measurements. The maximum measured error was 1.8 C, satisfying the +/-4 C requirement. The max-selection circuit was also tested with three distinct programmed inputs and repeated permutations; the output matched the selected worst-case input for all tested permutations.